

Code/Output

2. Write a program for process creation using C.

- Orphan Process

Code :

```
vaishnavi@DESKTOP-B9LM4V1: ~  
GNU nano 6.2 orphan.c *  
#include <stdio.h>  
#include <unistd.h>  
  
int main(){  
    pid_t pid = fork();  
    sleep(5);  
    printf("Child Process\n");  
    printf("PID: %d\n", getpid());  
    printf("PPID: %d\n", getppid());  
}  
    else{  
        printf("Parent Process exiting\n");  
    }  
    return 0;  
}bash: line 1: orphan.c: command not found  
bash: line 1: orphan.c: command not found
```

Output:

```
compilation terminated.  
vaishnavi@DESKTOP-B9LM4V1:~$ ls  
Zombie.save  orphan.c.save  orphan.c.save.2  zoombie  
orphan.c     orphan.c.save.1  zombie.c  
vaishnavi@DESKTOP-B9LM4V1:~$ nano orphan.c  
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan  
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan./orphan  
/usr/bin/ld: cannot open output file orphan./orphan: No such file or directory  
collect2: error: ld returned 1 exit status  
vaishnavi@DESKTOP-B9LM4V1:~$ gcc orphan.c -o orphan  
vaishnavi@DESKTOP-B9LM4V1:~$ ./orphan  
Parent Process exiting  
vaishnavi@DESKTOP-B9LM4V1:~$ cChild Process  
PID: 1645  
PPID: 1  
  
vaishnavi@DESKTOP-B9LM4V1:~$ ps -ef|grep orphan  
vaishna+ 1647 1519 0 01:19 tty1 00:00:00 grep --color=auto orphan  
vaishnavi@DESKTOP-B9LM4V1:~$
```

- **Zombie Process**

Code: -

```
Select vaishnavi@DESKTOP-B9LM4V1: ~
GNU nano 6.2 zombie.c
#include <stdio.h>
#include <unistd.h>

int main(){
    pid_t pid = fork();
    if(pid == 0){
        printf("Chlid process exiting\n");
    }else{
        sleep(30); // parent alive, no wait()
        printf("Parent process running\n");
    }
    return 0;
}
```

Output:

```
vaishnavi@DESKTOP-B9LM4V1: ~
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ Zombie.save
Zombie.save: command not found
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save orphan.c.save orphan.c.save.1 orphan.c.save.2 zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc Zombie.c -o zoombie
cc1: fatal error: Zombie.c: No such file or directory
compilation terminated.
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save orphan.c.save orphan.c.save.1 orphan.c.save.2 zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc zombie.c -o zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ ./zombie
-bash: ./zombie: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ nano zombie.c
vaishnavi@DESKTOP-B9LM4V1:~$ ls
Zombie.save orphan.c.save.1 zombie.c
orphan.c.save orphan.c.save.2 zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ gcc zombie.c -o zoombie
vaishnavi@DESKTOP-B9LM4V1:~$ ./zombie
-bash: ./zombie: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ ./zoombie
Chlid process exiting
Parent process running
vaishnavi@DESKTOP-B9LM4V1:~$ ps -el|grep z
vaishnavi@DESKTOP-B9LM4V1:~$ ps -el|grep Z
F S UID PID PPID C PRI NI ADDR SZ WCHAN TTY TIME CMD
vaishnavi@DESKTOP-B9LM4V1:~$
```

3. Create the process using fork 0 system call

- **Child Process creation**
- **Parent Process creation PPID and PI**

Code: -

```
Select vaishnavi@DESKTOP-B9LM4V1: ~
GNU nano 6.2 orphan.c *
#include <stdio.h>
#include <unistd.h>

int main(){
    pid_t pid = fork();

    if(pid == 0){
        printf("Chlid Process\n");
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());
    } else {
        printf("Parent Prpcess\n");
        printf("PID: %d\n", getpid());
        printf("Chlid PID: %d\n", Pid);
    }

    return 0;
}
```

Output: -

```
vaishnavi@DESKTOP-B9LM4V1:~$ nano fork.c
vaishnavi@DESKTOP-B9LM4V1:~$ ./fork
-bash: ./fork: No such file or directory
vaishnavi@DESKTOP-B9LM4V1:~$ nano fork.c
vaishnavi@DESKTOP-B9LM4V1:~$ gcc fork.c -o fork
vaishnavi@DESKTOP-B9LM4V1:~$ ./fork
Parent Process
Chlid Process
PID: 1666
PID: 1667
Chlid PID: 1667
PPID: 1666
vaishnavi@DESKTOP-B9LM4V1:~$
```